Hybrid LTD/CFF Representation for Repeated Propagators

Corrected master formula based on confluent residues and black-box numerator sampling

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Abstract

This note supersedes the previous implementation-oriented derivation. The earlier exported repeated-channel "hybrid" object was effectively a pure CFF object with a different orientation namespace, and the older local-chain prototype produced the desired-looking dotted-box orientation table but did not evaluate to the correct value when its JSON tree was used directly. The failure is conceptual: after repeated-copy localization, the equal-on-shell-energy limit is confluent. Individual local cone sectors contain singular factors whose cancellation is lost if they are exported as independent ordinary CFF products.

The corrected target formula is therefore not a relabelled CFF expression and not the old local-chain cone product. It is a confluent LTD/CFF interpolation formula. Repeated propagators are collapsed to powered channels for the denominator algebra, while numerator derivatives are avoided by exact multi-affine interpolation of the black-box numerator on a sector-dependent active set of edge energies. Each final orientation carries one concrete numerator replacement rule. In the absence of repeated propagators the active set is empty, all powers are one, and the formula collapses to ordinary LTD.

Contents

Chapter 1

Main Findings

1.1 What was wrong

Two tempting constructions are now ruled out.

CFF relabelling is not hybrid. The current repeated-channel fallback in the prototype builds the pure CFF bundle and relabels the orientations with a hybrid marker. This is numerically useful as a baseline, but it is not the hybrid formula. It has CFF orientations, CFF numerator substitutions, and CFF denominator trees.

The old local-chain tree is structurally plausible but incomplete. The older prototype emitted a dotted-box hybrid table with ordinary LTD-like skeleton orientations and repeated-copy signs. However, direct evaluation of that emitted tree gives values many orders of magnitude away from pure CFF and split-mass LTD. Its command-line test passed only because repeated-graph CFF and hybrid values were replaced by the smallest split-mass LTD proxy before reporting.

The missing ingredient is the confluent equal-energy limit. Repeated copies have the same spatial momentum and mass, so their on-shell energies are identical after the physical limit is taken. Local cone factors such as

$$\frac{1}{\omega_{r_1} - \omega_{r_2}}$$

are not meaningful as standalone JSON factors. They are distributions of a split-copy formula whose singularities cancel only after the full repeated sector has been recombined.

1.2 Correct strategy

The corrected construction has three parts.

1. Collapse every repeated set to one powered denominator channel.
2. Apply ordinary LTD residue logic to this collapsed graph, allowing higher-order residues when a powered channel is cut.
3. Before any higher-order residue can differentiate the black-box numerator, replace the numerator by an exact interpolation formula on the edge-energy variables that the higher-order residue would differentiate.

The final derivatives, or equivalently confluent divided differences, act only on known polynomials and known denominator factors. The black-box numerator is only sampled at explicit energy assignments.

Chapter 2

Input Data and Reduced Powered Graph

2.1 DOT graph and edge energies

An internal edge e carries momentum

$$q_e(k,p) = \sum_{r=1}^{L} \rho_{er} k_r + \sum_{a=1}^{n_{\text{ext}}} \eta_{ea} p_a, \qquad \rho_{er}, \eta_{ea} \in \mathbb{Z}, \tag{2.1}$$

and energy component

$$q_e^0(k^0, p^0) = \rho_e \cdot k^0 + \eta_e \cdot p^0. \tag{2.2}$$

For fixed spatial loop momenta and masses,

$$\omega_e = \sqrt{\vec{q}_e^{\,2} + m_e^2 - i0}. \tag{2.3}$$

The Feynman denominator is

$$D_e(q_e) = (q_e^0)^2 - \omega_e^2 + \mathrm{i}0 = (q_e^0 - \omega_e + \mathrm{i}0)(q_e^0 + \omega_e - \mathrm{i}0). \tag{2.4}$$

2.2 Repeated sets and collapse

A repeated set is an equivalence class

$$R_g = \{r_{g,1}, \dots, r_{g,\nu_g}\}, \qquad \nu_g \geq 2, \tag{2.5}$$

whose members have the same canonical routing up to orientation and the same symbolic mass key. Let

$$Q_g^0(k^0, p^0) = \rho_g \cdot k^0 + \eta_g \cdot p^0 \tag{2.6}$$

be the canonical channel energy and let

$$q^0_{r_{g,a}} = \epsilon_{g,a} Q_g^0, \qquad \epsilon_{g,a} \in \{+1, -1\}. \tag{2.7}$$

The denominator problem is reduced to a powered graph

$$\overline{E} = S \sqcup G_{\text{rep}}, \qquad S = E_{\text{int}} \setminus \bigcup_g R_g, \tag{2.8}$$

with powers

$$n_{\bar{e}} = \begin{cases} 1, & \bar{e} = s \in S, \\ \nu_g, & \bar{e} = g \in G_{\text{rep}}. \end{cases} \tag{2.9}$$

The collapsed denominator is

$$\prod_{\bar{e}\in\overline{E}} D_{\bar{e}}^{-n_{\bar{e}}} = \prod_{s\in S} D_s(q_s)^{-1} \prod_{g\in G_{\mathrm{rep}}} D_g(Q_g)^{-\nu_g}. \tag{2.10}$$

This collapse is only for denominator algebra. The numerator still knows about every original edge.

Chapter 3

Black-Box Numerator Interpolation

3.1 Allowed numerator class

The numerator is treated as a black box in edge-momentum representation. The assumption is:

> holding all other edge-energy variables fixed, the numerator is of degree at most one in each internal edge energy q_e^0.

This is the CFF/TOPT numerator condition. No symbolic derivatives of the numerator are allowed.

3.2 Why repeated-copy sampling alone is insufficient

Suppose a powered channel g is cut as a higher-order pole. The corresponding residue differentiates with respect to the channel energy Q_g^0. This derivative does not only see repeated-copy numerator variables. Any ordinary edge s with

$$\frac{\partial q_s^0}{\partial Q_g^0} \neq 0 \tag{3.1}$$

would also be differentiated. Therefore sampling only the repeated copies does not remove all numerator derivatives.

This is the key correction to the previous master formula.

3.3 Active edge set

For a collapsed LTD residue sector, let $B \subset \overline{E}$ be the chosen energy basis and let

$$H_B = \{g \in B \cap G_{\text{rep}} \mid \nu_g > 1\} \tag{3.2}$$

be the set of powered channels that are actually cut. These are the only sources of higher-order residue derivatives.

Define the active edge set $\mathcal{A}_B \subseteq E_{\text{int}}$ by

$$e \in \mathcal{A}_B \quad \Longleftrightarrow \quad q_e^0 \text{ depends on at least one cut powered-channel coordinate } Q_g^0,\ g \in H_B. \tag{3.3}$$

Equivalently, after changing to the energy basis containing B, e is active if q_e^0 has non-zero coefficient along any higher-order residue variable.

If $H_B = \varnothing$, then $\mathcal{A}_B = \varnothing$ and no interpolation is needed.

3.4 Exact multi-affine interpolation

For each active edge $e \in \mathcal{A}_B$ introduce a sample sign

$$\theta_e \in \{+1, -1\} \tag{3.4}$$

and the Lagrange factors

$$\ell_{e,\theta_e}(x) = \frac{1 + \theta_e x/\omega_e}{2}. \tag{3.5}$$

Since the numerator is multi-affine in active edge energies,

$$N_G(\{q_e^0\}_{e\in E_{\text{int}}}) = \sum_{\theta\in\{\pm1\}^{\mathcal{A}_B}} N_G\big(\{q_e^{0,\theta}\}\big) \prod_{e\in\mathcal{A}_B} \ell_{e,\theta_e}(q_e^0), \tag{3.6}$$

where

$$q_e^{0,\theta} = \begin{cases} \theta_e\omega_e, & e \in \mathcal{A}_B, \\ q_e^0, & e \notin \mathcal{A}_B. \end{cases} \tag{3.7}$$

The inactive edge energies remain ordinary LTD affine substitutions. In the final orientation, every edge receives a numerical replacement rule:

$$q_e^0(\mathfrak{o}) = \begin{cases} \theta_e\omega_e, & e \in \mathcal{A}_B, \\ \rho_e \cdot k_B^0 + \eta_e \cdot p^0, & e \notin \mathcal{A}_B. \end{cases} \tag{3.8}$$

Thus every orientation corresponds to one black-box numerator sample.

Chapter 4

Confluent LTD Residues

4.1 Ordinary sectors

If B cuts only simple channels, all $n_b = 1$. The residue is the ordinary LTD residue. The numerator is evaluated with the usual LTD affine edge-energy substitutions and no active interpolation:

$$\mathcal{A}_B = \varnothing.$$

4.2 Powered cut channels

If a cut channel $g \in B$ has power $\nu_g > 1$, the residue at

$$Q_g^0 = \sigma_g \omega_g, \qquad \sigma_g \in \{+1, -1\},$$

is confluent. In one variable,

$$\operatorname{Res}_{Q=\sigma\omega} \frac{F(Q)}{D(Q)^\nu} = \frac{1}{(\nu-1)!} \left(\frac{\partial}{\partial Q}\right)^{\nu-1} \frac{F(Q)}{(Q+\sigma\omega)^\nu}\bigg|_{Q=\sigma\omega}, \tag{4.1}$$

up to the standard LTD contour sign convention.

In the hybrid formula, $F(Q)$ is not allowed to contain a differentiated black-box numerator. Equation (3.6) is inserted first. The derivative in (4.1) then acts only on:

- known interpolation polynomials $\ell_{e,\theta_e}(q_e^0(Q))$;
- known uncut propagator factors;
- known Jacobian and opposite-energy factors.

The sampled numerator value $N_G(\{q_e^0(\mathfrak{o})\})$ is a constant for that orientation.

4.3 Derivative-free exported form

The mathematical derivation may use (4.1), but the exported JSON should not contain derivative operators. The known derivative is expanded algebraically into a finite sum

$$\mathcal{W}_{B,\sigma,\theta}(\vec{k}) = \sum_{\mu \in \mathcal{M}_{B,\sigma,\theta}} C_\mu(\vec{k}) \prod_{S \in \mathcal{S}_\mu} S(\vec{k})^{-p_{\mu,S}}. \tag{4.2}$$

Here S are causal LTD surfaces and $p_{\mu,S} \geq 1$ are integer powers. The multi-index μ records how the confluent derivative was distributed among known factors. If needed, μ is part of the orientation identifier even when it does not change the numerator replacement rule.

Chapter 5

CFF–Tau Interpretation of the Confluent Limit

5.1 Why the path-finder is not yet the physical organization

The implementation that currently passes the tests should be read as an algebraic path-finder. It proves that a derivative-free JSON expression with ordinary surface trees, repeated half-edge factors, and one numerator call per orientation can reproduce pure CFF and split-mass LTD for the tested affine numerator class. However, its organization is not yet the most physical one. The signs in a label such as

$$000+++|B3|b1|g1|nD0m$$

are collapsed-channel pole signs, not individual CFF signs of the repeated copies. Similarly, the present finite-stencil marker `nD0m` is an algebraic representation of a first derivative. In the current code this is implemented by a unit displacement in the residue-basis energy coordinate. That is numerically exact for affine numerators, but it is not dimensionally natural and should not be the final physics-facing formula.

The more physical target is to derive the same finite weights from the CFF energy integrations with localizing delta functions and their Fourier τ-parameters. In that organization every numerator substitution should be built from external energies and internal on-shell energies. No dimensionless constant energy shift should appear in the exported substitution maps.

5.2 Localizing repeated copies before taking the equal-energy limit

For one repeated channel with ν copies, introduce independent energies q_a^0 and localizing deltas,

$$\prod_{a=1}^{\nu} \frac{1}{D_a(q_a)} \prod_{a=1}^{\nu-1} \delta(q_{a+1}^0 - q_a^0). \tag{5.1}$$

Fourier expanding the deltas gives

$$\prod_{a=1}^{\nu-1} \delta(q_{a+1}^0 - q_a^0) = \int \prod_{a=1}^{\nu-1} \frac{\mathrm{d}\tau_a}{2\pi} \exp\left\{ \mathrm{i} \sum_{a=1}^{\nu-1} \tau_a (q_{a+1}^0 - q_a^0) \right\}. \tag{5.2}$$

For fixed τ-cone, each q_a^0 contour is closed according to the sign of the coefficient multiplying q_a^0 in the phase. This produces individual copy pole choices

$$q_a^0 = \sigma_a \omega_a, \qquad \sigma_a \in \{+1, -1\}, \tag{5.3}$$

together with a residual τ-cone orientation. These are the physically meaningful repeated-edge signs.

At this stage the ω_a should be regarded as distinct split energies. The CFF/tau expression is then a sum over ordinary simple-pole sectors. Its local weights contain factors depending on differences of signed energies, schematically

$$\frac{1}{\sigma_a\omega_a - \sigma_b\omega_b + \cdots}, \tag{5.4}$$

and the ellipsis denotes external-energy and skeleton-energy shifts generated by the residual graph. Individual sectors can be singular when the repeated copies are taken equal. The finite raised-propagator contribution is obtained only after summing a complete τ-cone family and then taking $\omega_a \to \omega_g$.

5.3 Tau cones become confluent divided differences

The equal-energy limit of the complete split-copy CFF/tau family is a confluent divided difference. This is the physical origin of the derivatives in (4.1). In one variable, a split expression of the form

$$\sum_{a=1}^{\nu} \frac{F(x_a)}{\prod_{b\neq a}(x_a - x_b)} \tag{5.5}$$

tends, as $x_a \to x$, to a derivative of order $\nu - 1$ acting on F. The LTD higher-order pole formula is therefore not an alien operation: it is the compact notation for the recombined CFF/tau sign-sector sum after localizing the repeated copies.

This observation also explains why the old local-chain table looked plausible but failed as a standalone JSON expression. It kept individual repeated-copy sign sectors, but it did not carry the complete singular divided-difference weight and did not perform the equal-energy recombination before export.

5.4 Physical numerator sampling

The numerator should remain black-box. The physical CFF/tau route replaces the unit finite stencil by on-shell samples.

1. Keep the repeated copies split through the local CFF/tau sign sum.
2. Evaluate the black-box numerator at the signed on-shell copy energies $\sigma_a\omega_a$ and at the affine skeleton substitutions generated by the same τ-cone.
3. Recombine complete τ-cone families into finite confluent weights.
4. Only then set $\omega_a = \omega_g$.

For an affine EMR-energy numerator this procedure must reduce to the same derivative information that the path-finder obtained from `nD0p`/`nD0m`. The difference is representational and physical: the samples are made at on-shell energies and external-energy shifts, rather than at arbitrary dimensionless displacements of a residue-basis coordinate.

The implemented affine reconstruction is expressed as moment matching. Let $\mathcal{S}$ be the physical sample orbit for the active repeated cut channels. A sample $s \in \mathcal{S}$ assigns a chain sign $\chi_g = \pm 1$ to each active collapsed channel and signs $\tau_{g,a} = \pm 1$ to its repeated copies, subject to the endpoint rule

$$\tau_{g,a} = +1\ \forall a \Rightarrow \chi_g = +1, \qquad \tau_{g,a} = -1\ \forall a \Rightarrow \chi_g = -1.$$

Mixed copy-sign sectors admit both chain signs. The loop energies are solved from $Q_g^0 = \chi_g \omega_g$, while repeated numerator edge energies are sampled directly at $q_{g,a}^0 = \tau_{g,a}\omega_{g,a}$. All substitutions are therefore linear combinations of external energies and internal on-shell energies; no dimensionful constant displacement appears.

For the value part of the confluent residue the rational weights c_s obey

$$\sum_{s\in\mathcal{S}} c_s = 1, \qquad \sum_{s\in\mathcal{S}} c_s q_{j,s}^0 = q_{j,\star}^0$$

for every loop and EMR edge energy q_j^0, where $\star$ is the collapsed diagonal pole. For one numerator derivative with respect to the active channel g, the exported term carries one extra $(2\omega_g)^{-1}$ half-edge factor and the weights obey

$$\sum_{s\in\mathcal{S}} c_s = 0, \qquad \sum_{s\in\mathcal{S}} c_s q_{j,s}^0 = 2\omega_g \frac{\partial q_j^0}{\partial Q_g^0}.$$

These equations are exact for every black-box numerator affine in the EMR energy variables.

When several repeated groups are active in the same residue, the implementation uses the affine compression of the physical orbit: the diagonal physical sample plus one complete sign orbit for each active group, keeping the other active groups on the diagonal. A full tensor-product orbit would only be needed for bilinear or higher numerator dependence between groups, which is outside the current CFF-compatible numerator class.

In the final JSON, such a recombined physical sector should still look like an ordinary orientation entry:

- the denominator tree stores the finite confluent surface powers;
- repeated `half_edges` store powers of cut 2ω factors;
- `pref` stores the rational/combinatorial weight;
- `loop_q0` and `edge_q0` store one physical numerator substitution rule.

Terms with the same numerator substitution rule are merged into one orientation; their denominator pieces are stored as top-level variants under that orientation. Thus the evaluator contract does not change: it still receives an orientation with one numerator map and a sum of ordinary denominator trees.

5.5 Implications for orientation labels

The current label

$$B3|b1|g1|nD0m$$

is useful for debugging the algebraic expansion:

- $B3$ means the collapsed powered channel is in the LTD basis;
- $b1$ means one confluent derivative was assigned to the numerator interpolation part;
- $g1$ means one derivative was assigned to the known denominator weight;
- `nD0m` meant the minus point of an algebraic numerator derivative stencil in the old path-finder.

The more physical label should record a recombined CFF/tau sector instead. A candidate structure is

$$(B, \sigma_{\text{chain}}; \Sigma_R; \mu),$$

where Σ_R denotes either a concrete repeated-copy sign pattern or, better, the complete τ-cone orbit whose equal-energy limit has already been recombined. For the dotted box, the current collapsed prefix $000+++$ should therefore not be interpreted as saying that the only physical repeated copy sector is $+++$. It says only that the collapsed third-order pole is the positive-energy pole of the repeated channel. Mixed repeated-copy sign sectors are expected to reappear in the physical CFF/tau organization as members of the orbit whose confluent limit produces the same $b/g/\mu$-type weights. In the current JSON labels this appears as a suffix such as

`physG3tpmpcm`,

meaning that the repeated group represented by edge 3 is sampled with copy signs $+, -, +$ and chain sign $-$. The old b/g markers may still appear inside merged denominator variants, but each exported orientation has a single physical numerator map.

5.6 Next theoretical target

The next derivation step should start from the split repeated-copy CFF/tau expression, not from the already-collapsed higher-pole residue. The concrete program is:

1. derive the local repeated-block τ-cone weights for a split ν-copy chain;
2. group sign sectors into complete orbits with finite equal-energy limits;
3. show that each orbit reproduces one or more terms in the current confluent expansion;
4. replace unit derivative stencils by on-shell divided differences so that every emitted energy substitution is a linear combination of external energies and internal on-shell energies;
5. keep the final output in the same orientation JSON format.

Chapter 6

Corrected Master Formula

6.1 Orientation data

A hybrid orientation is no longer determined by a single $\pm/0$ marker for each original edge. The correct orientation data are

$$\mathfrak{o} = (B, \sigma, \theta, \mu), \tag{6.1}$$

where:

- B is a rank-L collapsed LTD basis;
- σ chooses the pole sign for every cut channel in B;
- θ gives sample signs for the active edge set $\mathcal{A}_B$;
- μ selects one expanded confluent-weight term when powers are present.

The old edge-sign string can still be used as a compact prefix:

$$\varsigma_e = \begin{cases} \sigma_{\bar{e}}, & e \text{ represents a cut collapsed channel } \bar{e}, \\ 0, & \text{otherwise,} \end{cases}$$

but it is not a complete identifier for repeated graphs. A repeated graph may need suffixes such as active-edge sample signs and confluent multi-index labels.

6.2 Formula

Let $N_G^{\mathfrak{o}}$ denote the black-box numerator sampled with the replacement rule (3.8). The corrected hybrid spatial integrand is

$$I^{(3D)}_{G,\mathrm{hyb}}(\vec{k}) = \sum_{B\in\mathcal{B}} \sum_{\sigma} \sum_{\theta\in\{\pm1\}^{\mathcal{A}_B}} \sum_{\mu\in\mathcal{M}_{B,\sigma,\theta}} N_G^{\mathfrak{o}}(\vec{k})\, \mathcal{W}_{B,\sigma,\theta,\mu}(\vec{k}). \tag{6.2}$$

The weight $\mathcal{W}$ contains all half-edge factors, residue signs, interpolation-polynomial factors, and causal denominator surfaces, including any powers generated by confluent residues.

6.3 Collapse to ordinary LTD

If there are no repeated propagators, then every channel has power one. Therefore

$$H_B = \varnothing, \qquad \mathcal{A}_B = \varnothing, \qquad \mathcal{M}_{B,\sigma,\theta} = \{0\}.$$

The interpolation product is one, no confluent residue appears, and (6.2) reduces exactly to the ordinary LTD sum:

$$I^{(3D)}_{G,\mathrm{hyb}}(\vec{k}) = I^{(3D)}_{G,\mathrm{LTD}}(\vec{k}). \tag{6.3}$$

6.4 Relation to split-mass LTD

The split-mass test remains the most conservative numerical reference. Give the repeated copies distinct auxiliary masses, apply ordinary simple-pole LTD, and then take the equal-mass limit. The corrected formula is the analytic confluent limit of that split-mass LTD expression after the black-box numerator has been replaced by the interpolation identity (3.6). It should therefore agree with pure CFF and split-mass LTD, but it should not be structurally identical to either one for repeated graphs.

Chapter 7

Dotted Box Revisited

7.1 Why the 17-sector local-chain table is not enough

For the dotted box with ordinary edges $0, 1, 2$ and repeated copies $3, 4, 5$, the old local-chain output contained orientations such as

$$000 + - - \mid + \,.$$

This records repeated-copy signs and a chain sign, but it does not encode the confluent cancellation of equal repeated-copy energies. Direct JSON evaluation of that tree is therefore not a valid check of the formula.

7.2 Correct sector structure

Collapse edges $3, 4, 5$ to one powered channel g with power three. The collapsed one-loop graph has edges

$$\overline{E} = \{0, 1, 2, g\}, \qquad n_g = 3.$$

There are two qualitatively different sector types.

Ordinary cut sectors. If $B = \{0\}$, $B = \{1\}$, or $B = \{2\}$, no powered channel is cut. There is no higher-order residue and no active interpolation is required. The repeated channel appears as an uncut raised dual denominator evaluated on the ordinary LTD solution.

Repeated cut sectors. If $B = \{g\}$, the third-order pole at

$$Q_g^0 = \sigma_g \omega_g$$

is cut. The derivative order is two. The active set contains every original edge whose energy depends on Q_g^0, which in the usual box routing is all six internal edges:

$$\mathcal{A}_{\{g\}} = \{0, 1, 2, 3, 4, 5\}.$$

The numerator is sampled at

$$q_e^0 = \theta_e \omega_e, \qquad e = 0, \ldots, 5,$$

for the active-edge signs θ_e. The confluent derivative acts only on the six known interpolation factors and the known ordinary denominator factors.

This is more verbose than the old 17-sector table, but it is the price of avoiding numerator derivatives while keeping the numerator fully black-box. In the affine-numerator implementation this interpolation is emitted in the physical on-shell form described above: the numerator is sampled at signed on-shell copy energies and at the corresponding chain energy substitutions, and each distinct sample map is a separate JSON orientation.

Chapter 8

Implementation Consequences

8.1 JSON representation

The corrected formula can be stored in the same orientation JSON used by the pure CFF and pure LTD backends. No evaluator-side hidden backend is required. The needed ingredients are represented as follows.

- Surface powers S^{-p} are encoded by repeated surface identifiers in the existing factorization chain.
- Powers of cut half-edge factors $(2\omega_e)^{-p}$ are encoded by repeated entries in the existing `half_edges` list.
- Per-denominator rational prefactors from the expanded confluent weights are stored in the existing `pref` field.
- Known numerator-side energy factors are stored as repeated surface-cache references in `num_surfaces`. They multiply the sampled numerator instead of dividing the denominator.
- The surface cache is shared by denominator and numerator factors. If a numerator factor uses a linear form already present in a denominator tree, the same surface id is reused. A surface-cache entry carries `numerator_only=true` precisely when it appears only in `num_surfaces` and never in a denominator tree.
- One JSON orientation per distinct numerator energy map. The `orient_label` is a compact edge-map signature: $+$ and $-$ mean $q_e^0 = \pm\omega_e$, 0 means the numerator is sampled at $q_e^0 = 0$, and x means a non-trivial linear energy map shown in the detailed table.
- A `variants` list for all denominator pieces sharing that one numerator call. Physical origin labels such as CFF, LTD, pinch sectors, basis choices, β, and γ live on variants rather than in the orientation identity.
- Explicit physical-chain and repeated-copy sign metadata on the variant entries so tests can verify the numerator replacement rule.

For the currently supported numerator class, namely numerators affine in the edge-momentum-representation energy variables, numerator derivatives appearing in the derivation are replaced by exact physical on-shell interpolation weights. Each distinct edge-energy sample point is exported as one orientation with one concrete `edge_q0` substitution map. The `loop_q0` map is kept as derived auxiliary information for diagnostics and for the legacy `loops[...]` numerator hook, but the EMR numerator call is identified by `edge_q0`. The denominator derivative part is exported as a sum of ordinary surface products with repeated surface ids under `variants`. Thus the same evaluator consumes pure CFF, pure LTD, and hybrid CFF JSON, and the evaluator calls the black-box numerator once per orientation.

8.2 Python implementation map

The implementation follows the formula in a small number of modules.

- `graph_io.py` parses DOT graphs, detects repeated channels, and builds split-mass reference graphs.
- `orientation_bundle.py` constructs pure LTD residues, pure CFF contraction forests, the confluent hybrid interpolation bundle, and the bounded-degree CFF finite-pole contact sectors.
- `structure.py` serializes bundles to schema 6 JSON. This is where all terms with identical `edge_q0` maps are merged into one orientation and their denominator contributions become variants.
- `structure.py` also contains the JSON evaluator. It reads only the serialized surface cache, half-edge factors, numerator-surface factors, substitution maps, and factorization trees; no hidden CFF or LTD backend is consulted during evaluation.
- `pretty.py` renders the same JSON in audit form, showing numerator-only surfaces as (e) or (h), one row per denominator variant, and detailed substitution maps on request.

8.3 Bounded-degree CFF completion

The pure CFF builder also accepts optional per-edge bounds d_e on the power of each EMR edge energy in the numerator. These bounds are not needed for the ordinary affine tests, but they make it possible to decide whether higher powers can be represented by finite-pole data alone.

For each loop-energy variable k_i^0, the implementation computes the conservative one-variable degree estimate

$$\delta_i = \sum_{e:\,\partial q_e^0/\partial k_i^0 \neq 0} d_e - 2\,\#\{e : \partial q_e^0/\partial k_i^0 \neq 0\}.$$

The bounded-degree CFF completion is only emitted if every $\delta_i < -1$. Otherwise the command fails, because an infinity arc could contribute and no finite black-box sample formula is determined by the CFF pole data alone.

The recombination problem can be phrased algebraically. For each denominator channel define

$$D_e = (q_e^0)^2 - \omega_e^2.$$

Polynomial division by the finite-pole ideal generated by the D_e's gives a recursive decomposition

$$\frac{N}{\prod_e D_e} = \frac{R_e N}{D_e \prod_{f\neq e} D_f} + \frac{C_e N}{\prod_{f\neq e} D_f}, \qquad C_e N = \frac{N - R_e N}{D_e},$$

where $R_e N$ is the affine remainder in q_e^0. The first term keeps edge e and is ordinary CFF-safe in that variable. The second term cancels the propagator and is evaluated on the graph obtained by contracting edge e. Iterating (8.3) gives a pinch forest: each forest node is an ordinary CFF expression on a contracted subgraph. Hence all denominator surfaces can in principle be regular CFF E-surfaces; no LTD H-surface is intrinsic to the finite-pole completion.

For the special but important bound $d_e \leq 2$, this division is particularly simple. The quotient $C_e N$ is independent of q_e^0 and can be reconstructed from black-box samples by

$$C_e N = \frac{N(q_e^0 = +\omega_e) - 2N(q_e^0 = 0) + N(q_e^0 = -\omega_e)}{2\omega_e^2}.$$

For several quadratic edges the contact operators commute in the formal polynomial division, so a contact sector for a pinched set S carries the tensor product of the three-point weights

$$q_e^0 = +\omega_e,\ 0,\ -\omega_e, \qquad w_e = 1,\ -2,\ 1, \qquad e \in S,$$

together with one factor $1/(2\omega_e^2)$ per pinched edge. In the existing JSON this can be represented without new evaluator semantics by adding two `half_edges` entries for each pinched edge and multiplying the rational `pref` by $2w_e$. The denominator tree itself is the ordinary CFF tree of the contracted subgraph, lifted back to the original edge labels. Terminal one-denominator sectors are kept in the same JSON language by inserting a cached unit E-surface; the non-trivial factor is then only the half-edge Jacobian and the numerator sample. This is the first fully causal bounded-degree completion implemented in the prototype. The current implementation enables the one-loop non-repeated subcase with arbitrary numbers of quadratic-or-lower edge bounds. Multiloop pinches require a more careful choice of contracted CFF graph and loop-energy basis before this lifting is safe, and higher-order contacts require the numerator-side surface weights described below.

The same division also explains what is required for $d_e > 2$. Let $d = d_e$, choose $d - 1$ non-pole interpolation nodes

$$a_j \in \mathbb{Z}, \qquad a_j \neq \pm 1,$$

and let $L_j(x)$ be the Lagrange basis in $x = q_e^0/\omega_e$. Then the contact quotient can be reconstructed as

$$C_e N(q_e^0) = \sum_j \frac{L_j(q_e^0/\omega_e)}{(a_j^2 - 1)\omega_e^2} \left[N(a_j\omega_e) - \frac{a_j + 1}{2} N(+\omega_e) - \frac{1 - a_j}{2} N(-\omega_e) \right].$$

For a cubic term, for example,

$$\frac{(q_e^0)^3 - R_e(q_e^0)^3}{D_e} = q_e^0,$$

so an isolated cubic contact produces only a known affine numerator-side factor, which the current CFF lift can handle. For quartic and for mixed cubic sectors the weights become quadratic or higher polynomials in the remaining edge energies. Those known factors must be reduced recursively by the same finite-pole division before ordinary CFF is valid on the contracted graph.

The implementation therefore deliberately emits only the cases that are currently causal and tested:

- one-loop non-repeated graphs with arbitrary quadratic-or-lower caps;
- isolated single-edge cubic caps;
- no LTD-contact fallback.

Unsupported higher contacts, multiloop pinches, and repeated-channel bounded hybrid sectors raise an implementation error instead of serializing CFF + (LTD − CFF). This is intentional: that fallback is a useful numerical identity but not the desired E-surface-only CFF representation.

8.3.1 Status of the higher-power lift

The obstruction to arbitrary capped EMR energy powers is not the one-dimensional finite-pole interpolation itself. The interpolation formula (8.3) exactly reconstructs the quotient by $(q_e^0)^2 - \omega_e^2$ for a bounded polynomial in one edge energy. The difficulty is what the quotient contains after the edge has been pinched. For degree $d_e > 2$, the contact term carries known polynomial factors in the remaining edge-energy variables. Those factors are not black-box numerator samples, but they are still numerator-side energy dependence. An ordinary CFF tree on the pinched graph is valid only after these known factors have also been reduced by the same finite-pole division whenever they exceed the affine-safe class.

Consequently, a general capped-power completion requires an internal sector algebra rather than a single extra CFF branch. Each sector must track

- the black-box numerator sample map;
- the known numerator-side polynomial weight, expressed in linear energy forms and cached surfaces;
- the denominator graph or minor on which the next CFF tree is built;
- the remaining degree bounds and the loop-energy UV degree of the sector;
- the merge key used to combine all denominator variants with the same numerator energy map.

Only after this recursive reduction terminates can the sector be serialized as ordinary JSON using denominator E-surfaces and numerator-side surface factors. This is the path for arbitrary caps in pure CFF. The same sector algebra is also the natural implementation layer for the hybrid formula, where finite-pole contacts must be combined with the confluent repeated-channel residue weights.

The quadratic case $d_e \leq 2$ is special and substantially easier. The quotient (8.3) is independent of q_e^0, so no known polynomial factor is produced by a quadratic contact. Arbitrary combinations of quadratic-powered edges should therefore be attainable without the recursive polynomial-weight machinery. What remains is the correct multiloop contact operation: after several edges are pinched, the CFF denominator tree must be built on the right causal minor, with a loop-energy basis for which the finite part has no residue at infinity. Naive contraction and naive deletion are not equivalent to this operation in general. This is why the current public builder keeps the multiloop guard even though the one-loop quadratic formula is already implemented and tested.

The intended implementation order is therefore:

1. complete the pure-CFF quadratic contact lift for arbitrary convergent combinations of quadratic edges, validating every sector directly against LTD;
2. extend the same quadratic contact operator to repeated-channel hybrid sectors, preserving the collapse to LTD when no repeated propagator is present;
3. add the recursive known-polynomial sector algebra needed for cubic, quartic, and mixed higher-power caps.

8.3.2 Higher powers on repeated channels

The same numerator-surface mechanism should also extend the repeated-channel part of the hybrid master formula. Let a repeated group g have collapsed denominator power ν_g and channel energy Q_g^0. If the numerator has a capped polynomial dependence on Q_g^0, divide it recursively by

$$D_g = (Q_g^0)^2 - \omega_g^2.$$

Equivalently,

$$N = R_0 + D_g R_1 + D_g^2 R_2 + \cdots , \qquad \deg_{Q_g^0} R_r \leq 1.$$

Inserted against $D_g^{-\nu_g}$, the term $D_g^r R_r$ reduces the repeated power to $D_g^{-(\nu_g - r)}$ while $r < \nu_g$. If all copies are cancelled, the corresponding channel is pinched out of the collapsed graph. Thus high numerator powers on a repeated channel become a finite sum of lower-power hybrid sectors plus known numerator-side energy factors.

This is exactly what is needed to preserve the black-box numerator contract. The confluent residue of a cut repeated channel differentiates only the known factors generated by the division and the known causal denominators. The remainder R_r, being affine in the active channel energy, is reconstructed from finitely many black-box samples as in the current hybrid interpolation. For multivariate EMR numerators the division is performed in the ideal generated by the active channel denominators D_g. The quotient weights can contain mixed-sign linear energy forms; these are acceptable as numerator-only H-class surface-cache entries because they are polynomial multipliers, not poles.

The expected fully causal high-degree hybrid form is therefore

$$\sum_{\text{pinch forests } F} \mathrm{CFF}_{\text{hybrid}}(G/F, \{\nu_g - r_g\}) \times \prod_{a \in A_F} S_a,$$

where every denominator tree is built from ordinary CFF E-surfaces of the appropriate collapsed or pinched graph, and each S_a is a cached numerator-side surface id. The remaining implementation work is mechanical but nontrivial: choose a deterministic multivariate division order, emit unit denominator sectors when all powers of a channel cancel, and control the growth of interpolation samples from the user-provided degree caps.

8.4 Testing requirements

For any repeated graph, a valid three-way test must check both numerical agreement and structural independence:

- pure CFF, hybrid, and split-mass LTD values must agree within tolerance;
- none of the three reported values should be exactly the same string or the same exact high-precision object, except for graphs with no repeated propagators;
- hybrid JSON must contain LTD-collapse sectors, active-edge sampled sectors, or confluent surface powers when appropriate;
- repeated hybrid JSON must not contain two orientations with the same numerator energy map;
- physical numerator maps must have no non-zero constant energy shifts;
- hybrid must collapse structurally to LTD when no repeated group is present.

For bounded-degree CFF tests, the suite checks that quadratic or cubic EMR energy numerators agree against LTD on split-mass versions of the existing one-loop, multiloop, and five-loop examples, and that deliberately divergent degree bounds are rejected before evaluation.

8.5 Current prototype status

The repeated-channel Python backend now implements the affine-numerator confluent version described above. It collapses repeated copies to powered channels for the denominator algebra, expands the confluent derivatives into ordinary denominator variants, and samples the black-box numerator at explicit physical on-shell EMR-energy assignments. The implementation deliberately no longer relabels pure CFF orientations as hybrid orientations.

Two limitations remain explicit. First, the physical interpolation currently assumes the numerator is affine in the EMR energy variables; this covers the edge–external dot-product tests used for the repeated topologies. Second, the old local-chain implementation remains useful only as a structural warning: its orientation table is not a correct evaluable master formula by itself.

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